

INCOSE GtWR 4.0

Implementation of the latest concepts

“At the crossroads between needs and requirement statements”



Ilyes Yousfi

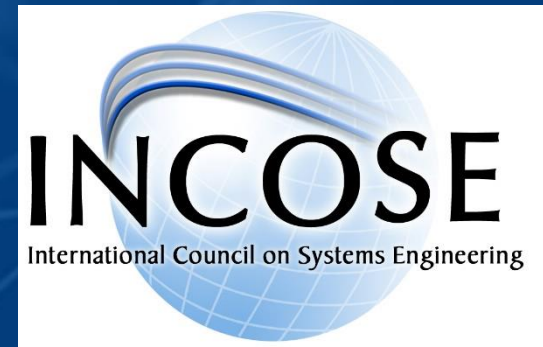
The REUSE Company

ilyes.yousfi@reusecompany.com

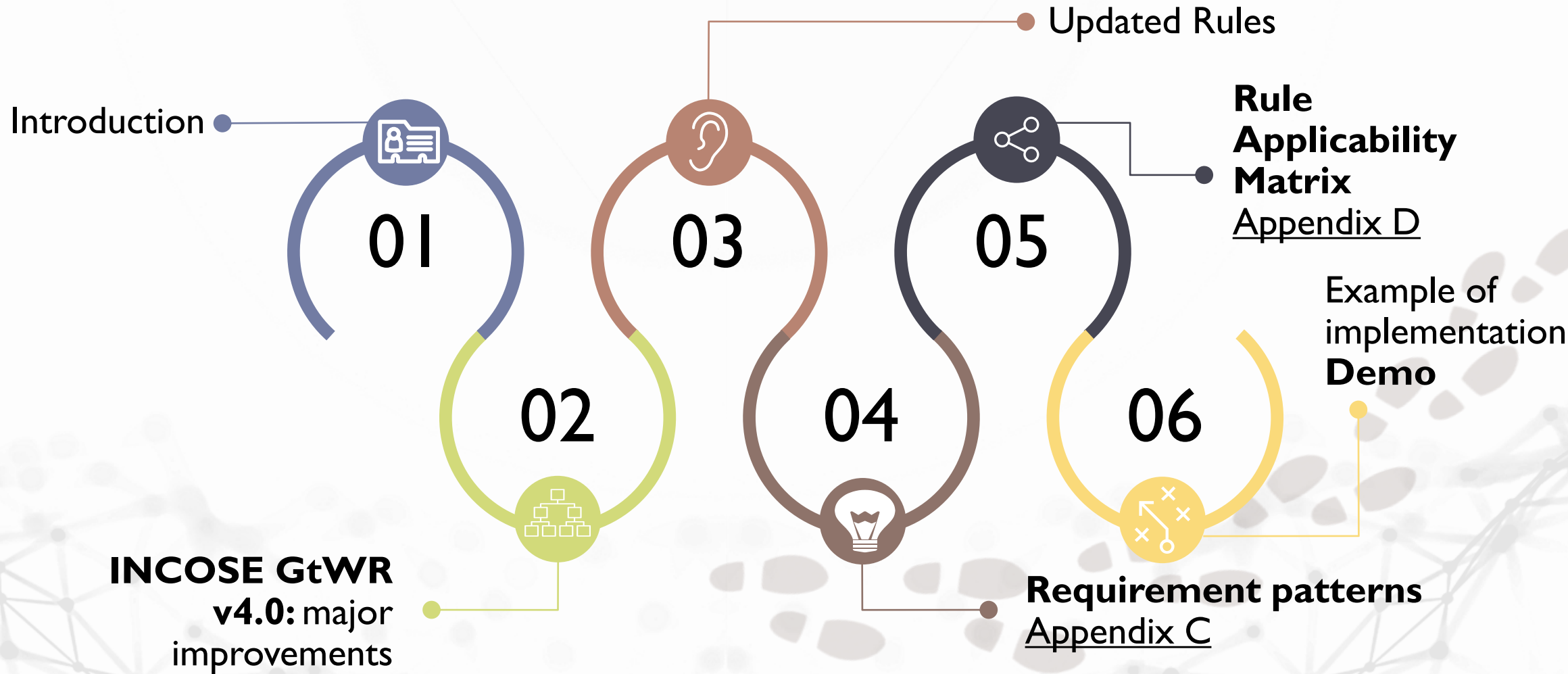


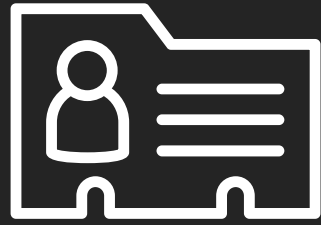
THE
REUSE
COMPANY

September 19th, 2023



Contents





1. Brief intro

The REUSE Company

02

Systems + Software Engineers

Smart combination between
Company staff and R&D (University)

04

To promote **reusable** solutions to
SMART and Systems Engineering
environments, by offering a
knowledge centric approach.

**WHEN**

01

The company was
established in **1999**

Spin-off of a University in
Madrid

WHO**WHERE**

03

Headquarters:
Madrid (Spain)

International offices:
Stockholm (Sweden)
Tokyo (Japan) Delegation

WHY

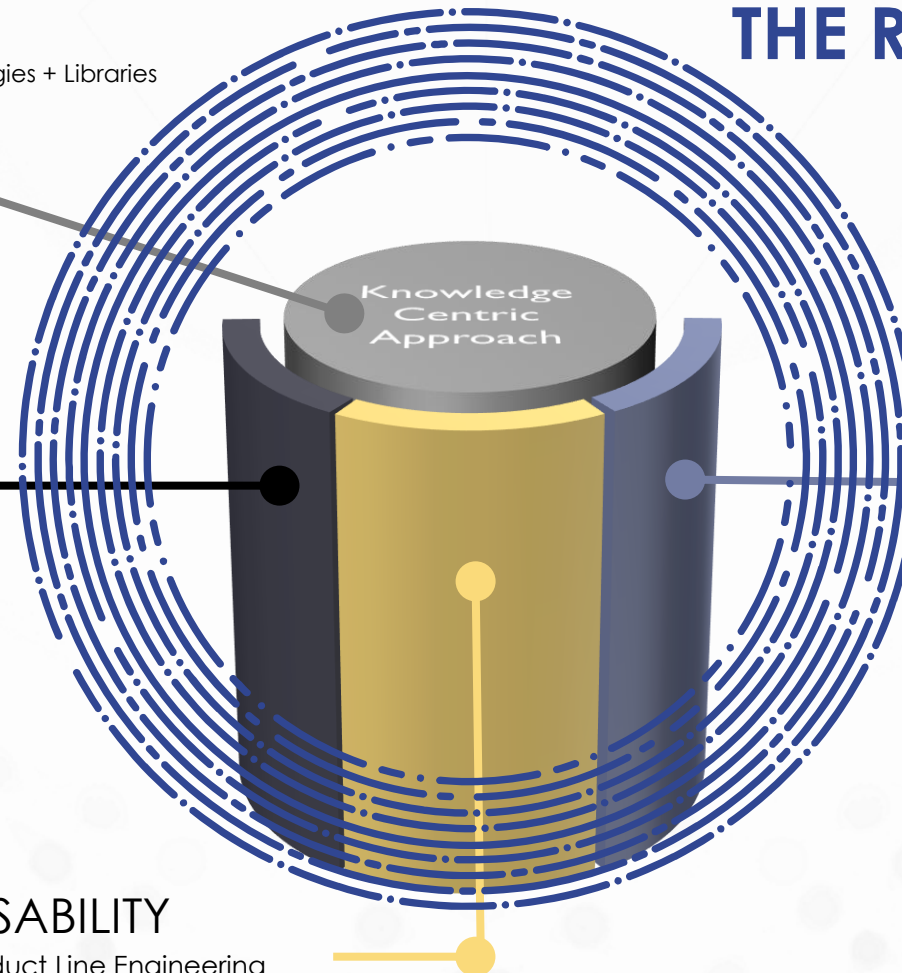
Knowledge Centric Systems Engineering (KCSE)

Connectivity & Interoperability
Configuration Management
SMART Authoring
Lifecycle Management
Global Repositories: Ontologies + Libraries

THE REUSE COMPANY = TRC

TRACEABILITY

Links and Interoperability –
Transformations –
Change management –



QUALITY

- Quality Management
- Verification & Validation management
- Alerts & Risks Management
- Decision management



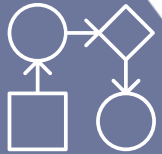
REUSABILITY

- Product Line Engineering
- Retrieval & Archiving
- Adaptability to Existing Toolsets



THE REUSE COMPANY = TRC

01



SES TMP –
TMx

Technical Management
Processes:
Quality
V&V
Traces...

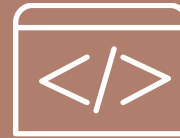
02



SES Interop –
HUBx

Interoperability HUB
Connectivity to
external sources

03



SES Req Eng –
REx

SMART Requirements
Engineering
Using MS Word as
tech platform

04



SES SLCM –
SLx

System Life cycle
management

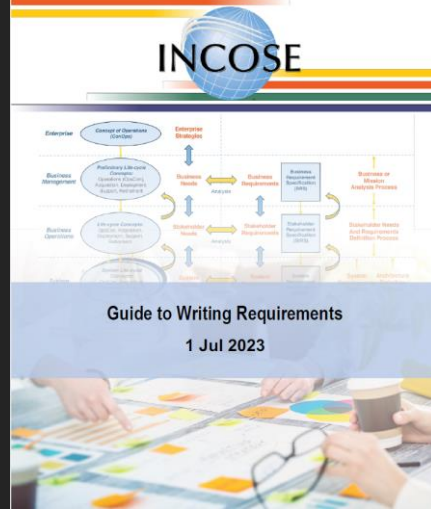
05



Knowledge –
KMx

Knowledge and Information
Management
Ontologies
Virtual Assistants
Search eng.

The REUSE Company is specialized
in the application of
semantic technologies and
artificial intelligence to improve
the **digitalization** of
the **Systems Engineering life cycle**.



2. INCOSE GtWR v4.0

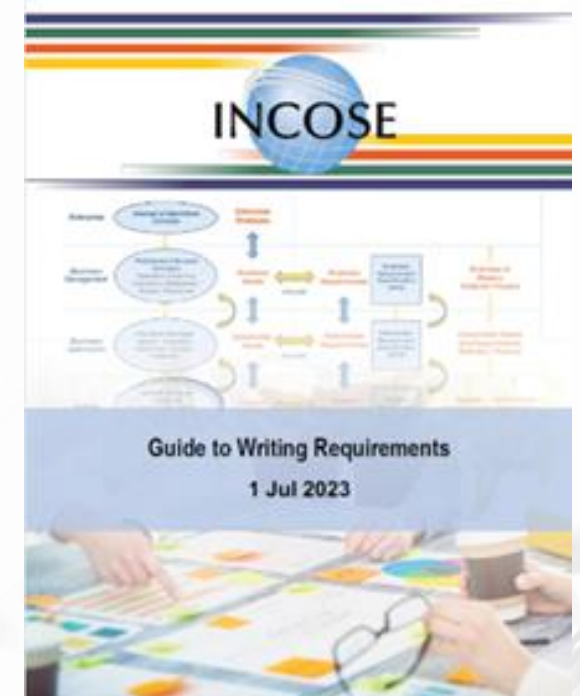
**Major
improvements**

INCOSE Guide for Writing Requirements: get to know it!

The INCOSE RGW (**Requirements Working Group**) in line with its goal
(Expand and promote the body of knowledge of requirements engineering and its benefits within the systems engineering community)
has developed the INCOSE GtWR (**Guide to Writing Requirements**)

The GtWR provides guidance on how to express textual
needs AND requirements expressions.

The GtWR draw advice into a single, comprehensive set of
characteristics, rules and attributes
for well-formed need and requirement expressions.



INCOSE GtWR v4.0

➤ **15 Characteristics** of individual and sets of needs and requirements.

➤ **42 Rules** to help formulate individual and sets of needs and requirements.

➤ **49 Attributes** are attached to need or requirement statements to form need or requirement expressions.

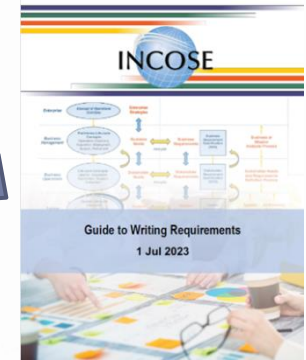
42 Rules / 15 Characteristics

Type	Rule Number	Rule name	C1 - NECESSARY	C2 - APPROPRIATE	C3 - UNAMBIGUOUS	C4 - COMPLETE	C5 - SINGULAR	C6 - FEASIBLE	C7 - VERIFIABLE	C8 - CORRECT	C9 - CONFORMING	C10 - COMPLETE	C11 - CONSISTENT	C12 - FEASIBLE	C13 - COMPREHENSIBLE	C14 - ABLE TO BE VALIDATED	C15 - CORRECT
Accuracy	R01	Sentence Structure		1	1			1	1	1							
	R02	Use Active Voice		1	1	1											
	R03	Subject Verb		1	1			1				1			1		
	R04	Use Defined Terms			1			1				1		1	1	1	
	R05	Use Definite Articles			1			1									
	R06	Units			1	1		1	1								
	R07	Avoid Vague Terms			1	1		1									
	R08	No Escape Clauses			1	1		1									
	R09	No Open Ended			1	1	1	1									
	R10	Superfluous Infinitives			1			1									
Concision	R11	Separate Clauses			1	1		1	1								
	R12	Correct Grammar			1			1	1	1							
Non Ambiguity	R13	Correct Spelling			1			1									
	R14	Correct Punctuation			1			1									
	R15	Logical Condition			1			1									
	R16	Avoid Not			1			1	1								
	R17	Oblique			1			1									
Singularity	R18	Single Sentence			1		1	1		1					1		
	R19	Avoid Combinators			1		1										
	R20	Avoid Purpose		1			1										
	R21	Avoid Parentheses					1										
	R22	Enumeration			1		1										
Completeness	R23	Context			1	1	1										
	R24	Avoid Pronouns			1	1		1									
	R25	Use Of Headings			1												
Realism	R26	Avoid Absolutes						1	1	1			1				
Conditions	R27	Explicit				1		1	1								
	R28	Explicit Lists			1			1									
Uniqueness	R29	Classify										1	1				
	R30	Express Once		1							1		1				
Abstraction	R31	Solutionfree		1													
Quantifiers	R32	Universals			1			1	1								
Tolerance	R33	Value Range			1	1		1	1				1				
Quantification	R34	Measurable			1	1		1						1			
	R35	Temporal Indefinite			1	1		1									
Uniform Language	R36	Use Consistent Terms			1					1	1		1	1	1	1	
	R37	Define Acronyms			1					1		1	1	1	1	1	
	R38	Avoid Abbreviations								1		1	1	1	1	1	
	R39	Style Guide				1	1			1		1	1	1	1	1	
	R40	Decimal Format				1	1			1		1					
Modularity	R41	Related Requirements				1				1	1	1	1	1	1	1	
	R42	Structured										1	1		1	1	1



49 Attributes

Attribute	Attributes to Help Define the Requirement and its Intent	Associated with the System of Interest (SOI) Verification	Attributes to Help Maintain the Requirements	Attributes to Show Applicability and Allow Reuse
A01: Rationale*	1			
A02: SOI Primary Verification or Validation Method*	1			
A03: SOI Verification or Validation Approach	1			
A04: Trace to Parent*	1			
A05: Trace to Source*	1			
A06: Condition of Use	1			
A07: States and Modes	1			
A08: Allocation*	1			
A09: SOI Verification or Validation Level		1		
A10: SOI Verification or Validation Phase		1		
A11: SOI Verification or Validation Results		1		
A12: SOI Verification or Validation Status		1		
A13: Unique Identifier*			1	
A14: Unique Name			1	
A15: Originator/Author*			1	
A16: Date Requirement Entered			1	
A17: Owner*			1	
A18: Stakeholders			1	
A19: Change Board			1	
A20: Change Status			1	
A21: Version Number			1	
A22: Approval Date			1	
A23: Date of Last Change			1	
A24: Stability			1	
A25: Responsible Person			1	
A26: Need or Requirement Verification Status*			1	
A27: Need or Requirement Validation Status*			1	
A28: Status (of the Need or Requirement)*			1	
A29: Status (of Implementation)			1	
A30: Trace to Interface Definition			1	
A31: Trace to Peer Requirements			1	
A32: Priority*			1	
A33: Criticality or Essentiality*			1	
A34: Risk (of Implementation)*			1	
A35: Risk (Mitigation)			1	
A36: Key Driving Need or Requirement (KDN/KDR)			1	
A37: Additional Comments			1	
A38: Type/Category			1	
A39: Applicability				1
A40: Region				1
A41: Country				1
A42: State/Province				1
A43: Application				1
A44: Market Segment				1
A45: Business Unit				1
A46: Business (Product) Line				1



INCOSE GtWR: more details about the updates

Overview of the INCOSE GtWR v4.0

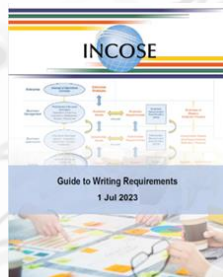
Guide to Writing Requirements Version 4 Overview

Lou Wheatcraft, Senior Consultant, Wheatland Consulting
Wheatland.consulting@gmail.com

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- ❖ Change summary
- ❖ Details about the updated definitions and concepts
- ❖ Added figures in Elaboration section of Characteristics & Rules

[Link](#) (INCOSE RWG Youtube Channel)



INCOSE GtWR v4.0 release: focus of this presentation

i) Rules:

- Updated Rules: R06 (Units), R33 (Tolerance)
- ****New Rule**** : R40 (DecimalFormat)

ii) Requirement Patterns (Appendix C)

iii) Rule Applicability Matrix (Appendix D) : Needs vs. Requirements and Data Dictionaries

THE CHALLENGE

“address both need and requirement statements and understand the differences between the two”

INCOSE GtWR sect. 1.8



3. Rules

Updates in v4.0 release

➤ Updated Rule

➤ Rule 06/ Common Units of Measure

Definition:

When stating quantities, all numbers should have appropriate and consistent units of measure explicitly stated using a common measurement system in terms of the thing the number refers.

- Added the triplet {*Component, Property, MeasurementUnit*} consistency concept

Example:

- REQ-001 "The Fuel_System shall have a maximum Fuel_Tank volume of 60 l." → {Fuel_System, Fuel_Tank volume, l}
- REQ-002 "When the Sensor detects the Fuel_Tank volume is less than 50 dl, the Sensor shall light the Low-Fuel_indicator within 1 second." ❌
- REQ-003 "While the Fuel_System has a Fuel_Tank volume higher than 60 l, the Low-Fuel_indicator light shall be Off." ✅
- REQ-004 "While the engine is on, If the Cooling_System_volume is lower than 3400 ml, the Water_Pump shall transit to Emergency mode." ✅

Updates in v4.0 release

➤ Updated Rule

➤ Rule 33 / Range of Values (Tolerance)

Definition:

Define each quantity with a range of values appropriate to the entity to which the quantity applies and against which the entity will be verified or validated.

➤ Uniformity of Format

➤ Use a consistent format for tolerances

Example:

REQ-004 "The Fuel_System shall have a maximum weight of 600 lb $\pm 5\%$."

REQ-005 "The Fuel_System shall have a maximum power consumption of 2.4 kW $(-0.0 \text{ kW} / +0.4 \text{ kW})$ "

➤ Unit for each value?

➤ Numerical vs. textual format : "between...and..." / "from....to..." etc.



Updates in v4.0 release

➤ New Rule

➤ Rule 40 / DecimalFormat

Definition:

Use a consistent format and number of signification digits for the specification of decimal numbers.

➤ Symbol for decimals

’ VS. •

➤ ...and symbol for thousands !

➤ Set a convention

Example:

	’	•
	9,499	9.499
•	9,499.99	’ 9.499,99

Updates in v4.0 release

➤ New Rule

- Rule 40 / DecimalFormat
- Decimal format for numbers between -1 and 1
 - ,99 vs. **0,99**



4. Requirements patterns

INCOSE GtWR appx. C

Benefits of requirement patterns

- Sets an internal standard for well-formed requirements
- Help identifying keywords (i.e. the subject before the “shall”)
- Customizable for every domain, customer and content of each requirements document
- **Pattern Libraries**

Condition Keyword

Condition

Agent

Shall

Action

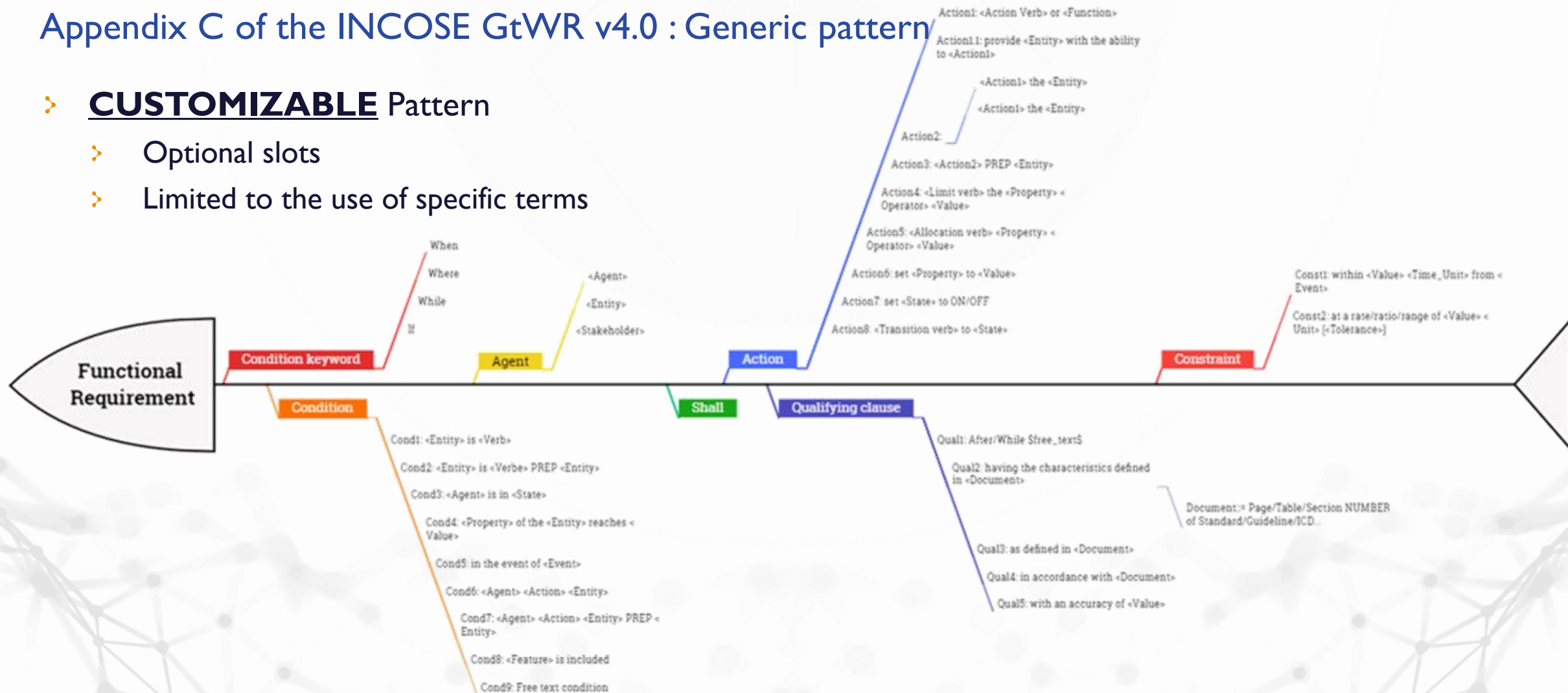
Qualifying Clause

Constraint

Appendix C of the INCOSE GtWR v4.0 : Generic pattern

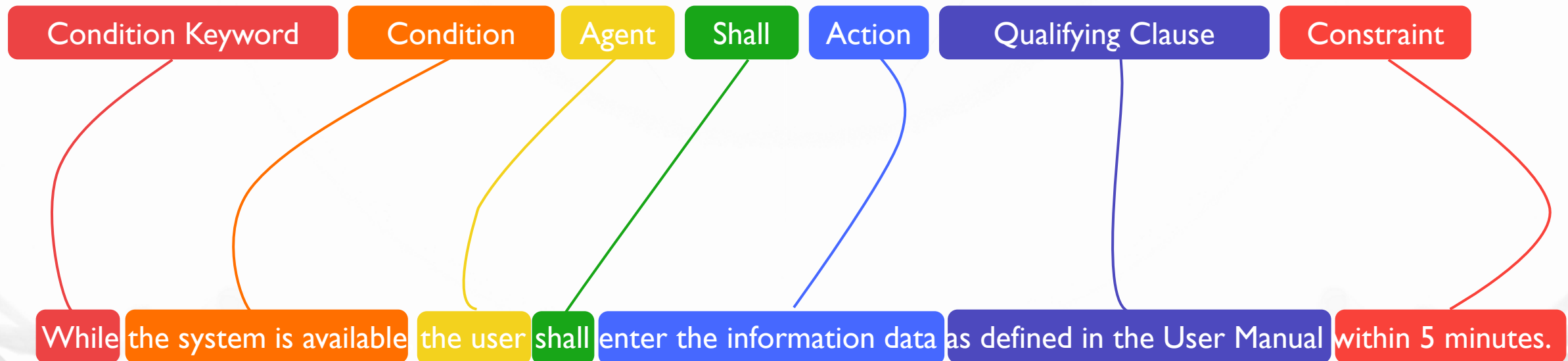
➤ **CUSTOMIZABLE** Pattern

- Optional slots
- Limited to the use of specific terms



Appendix C of the INCOSE GtWR v4.0 : Generic pattern

➤ **CUSTOMIZABLE** Pattern



Appendix C: Example of slot restriction

➤ Importance of a Defined Data Dictionary

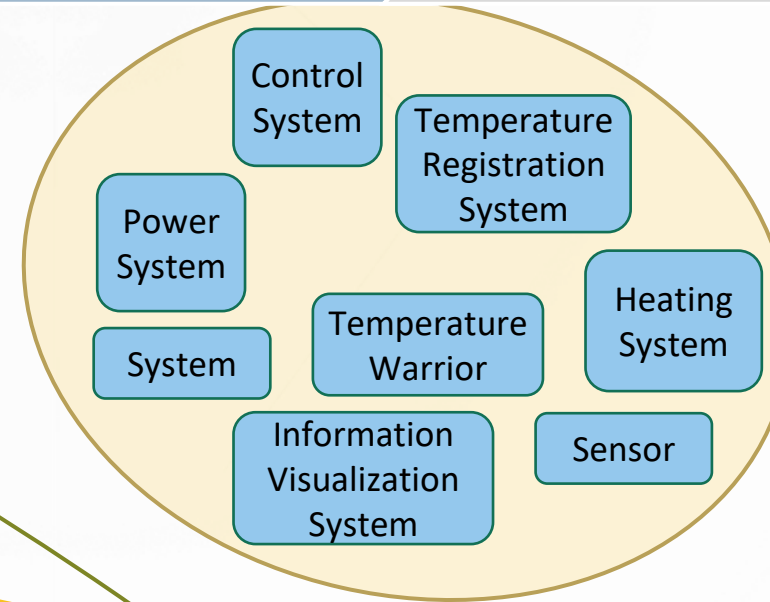
Condition

<<AGENT>> in <<STATE>>



Examples matching this part of the pattern:

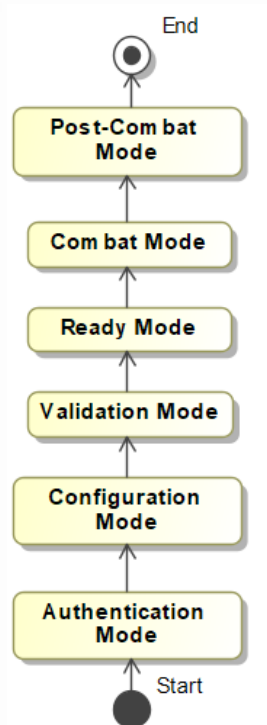
- “Control System in Ready Mode”...
- “Temperature Warrior in Combat Mode”...
- “Information Visualization System in Configuration Mode”...



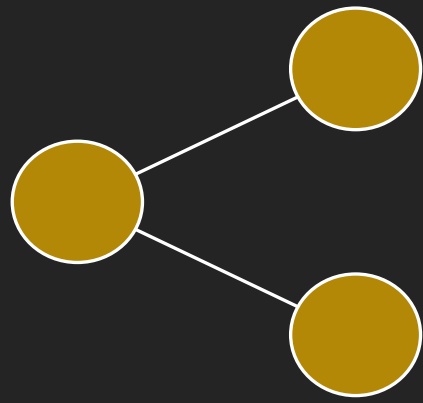
<<AGENT>>



Data dictionary



<<STATE>>



5. Rule Applicability Matrix (Appendix D)

Introduction to the Rule Applicability Matrix

➤ Guidelines/Recommendations to apply a tailored set of rules to either Needs or Requirements

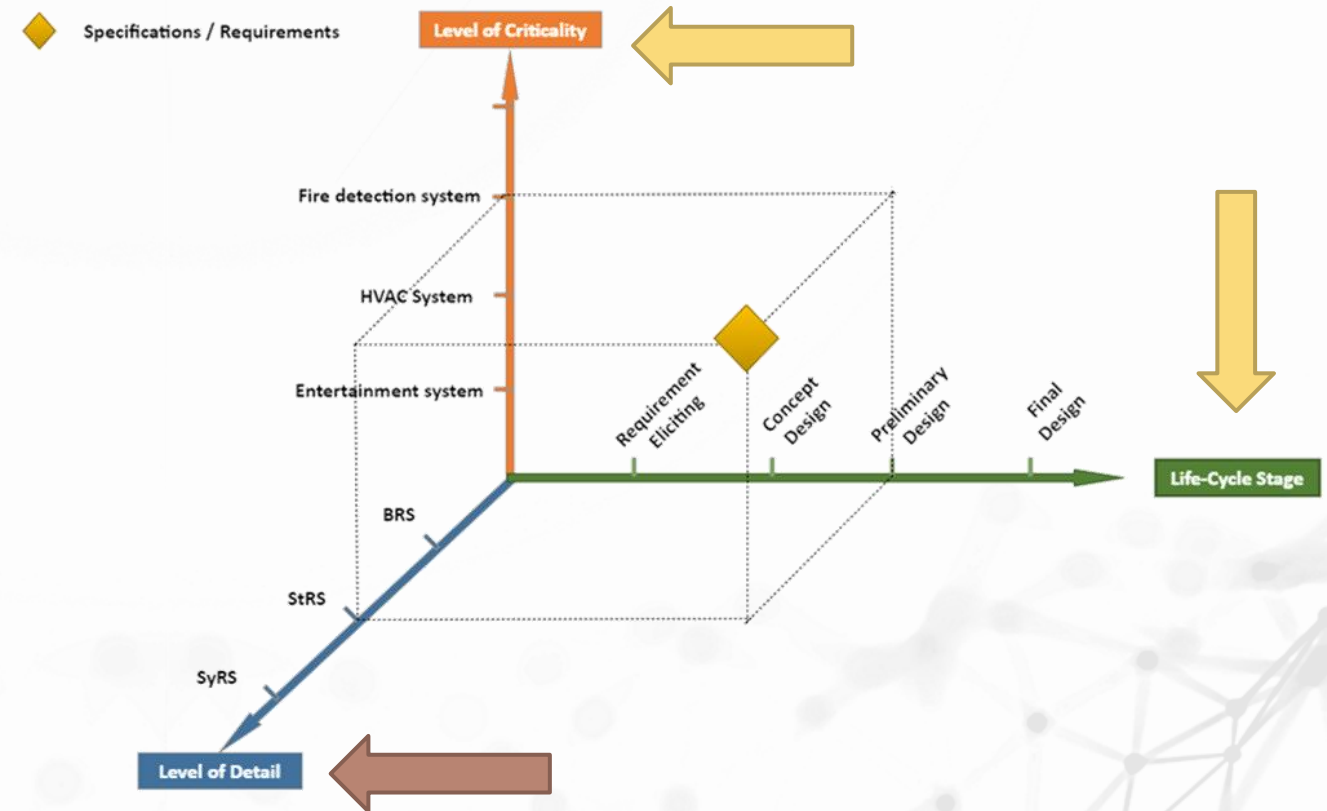
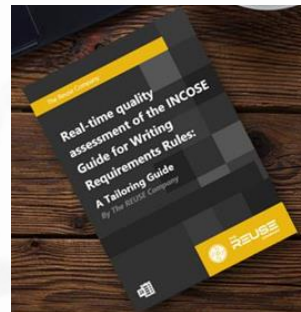
- *Optional / Recommended* rule for Needs
- *Compulsory* rules for Requirements

➤ **Tailoring** to either Needs or Reqs

➤ Is a **Project Data Dictionary** needed?

- 1. check cross-domain aspects
- 2. check domain-specific aspects

Tailoring Guide
(Link to download the book) :



Needs & Requirements – what is the difference?

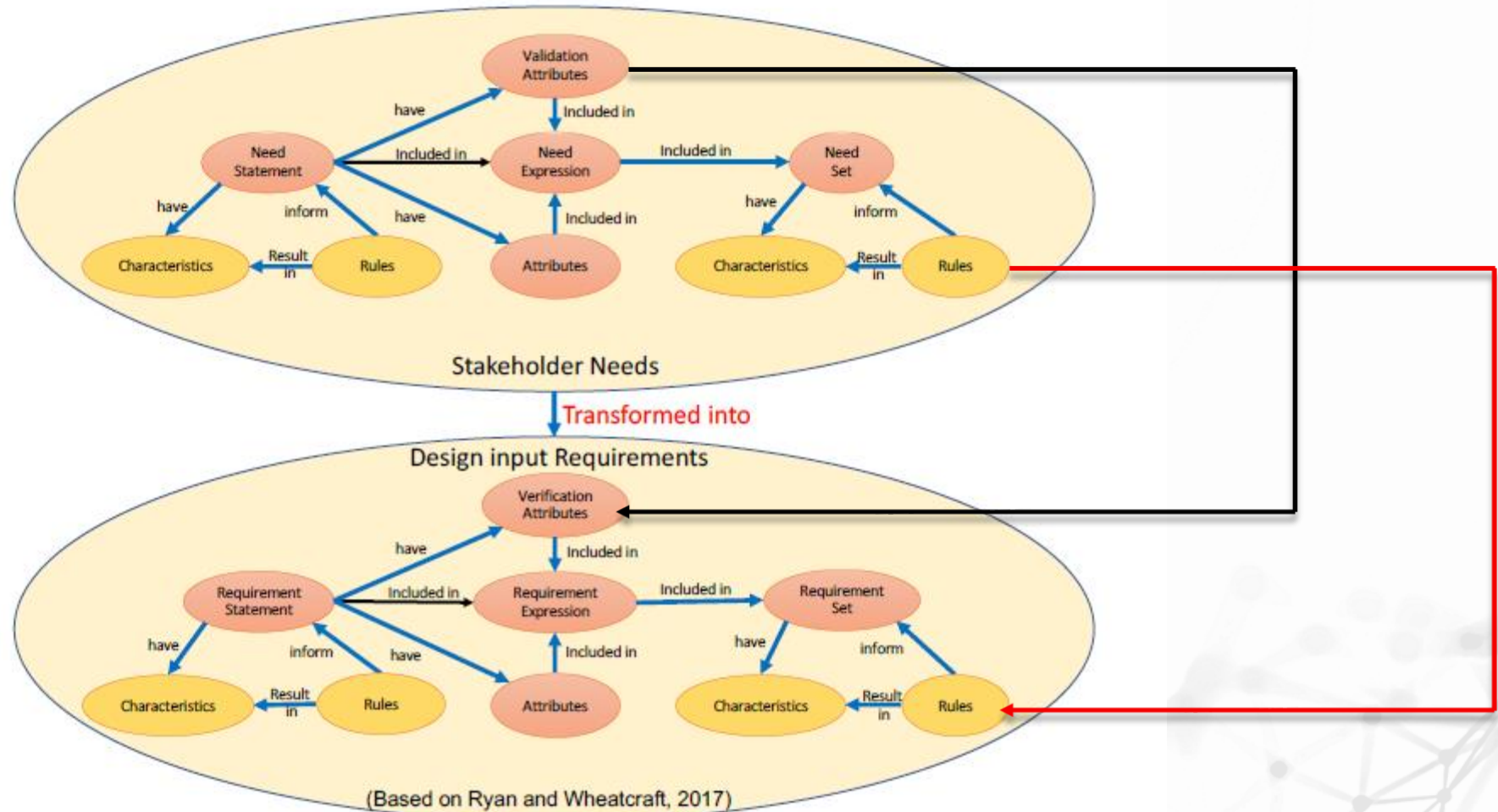


Figure 4: Entity-Relationship Diagram for Needs and Requirements Terms.

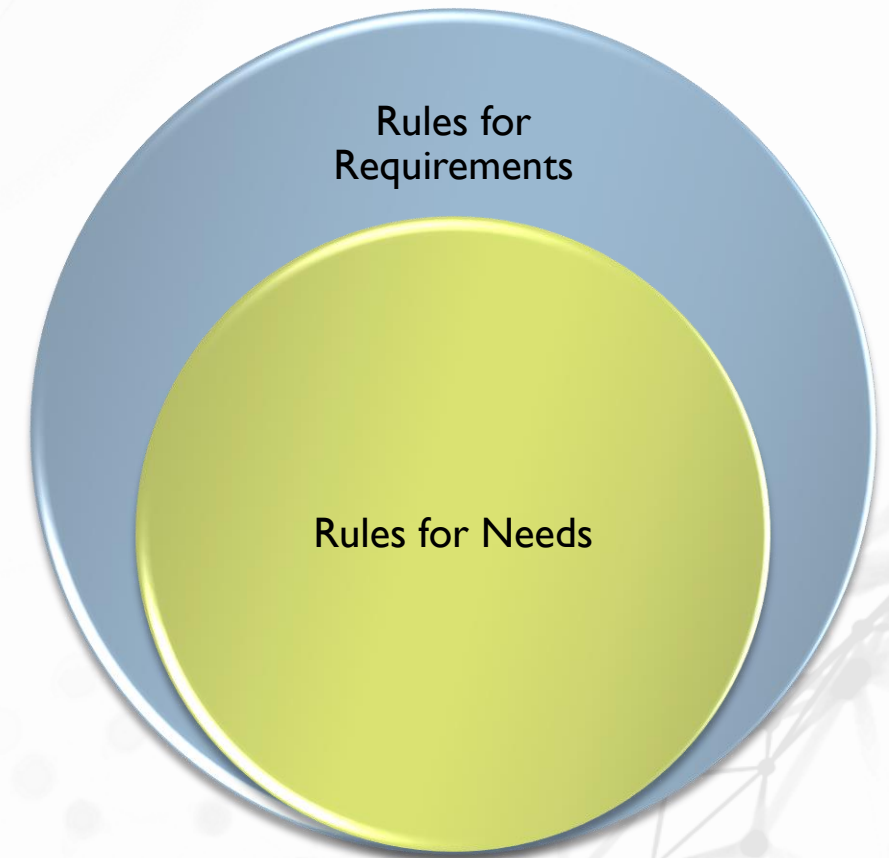
Characteristics of Needs and Requirements

- Different purposes
 - Needs -> Meet the business expectations
 - Requirements -> What to do to achieve the needs.

But...

Requirements translate needs ('Derived Requirements')

- Rules that apply to needs would also apply to requirements.
- I Need -> X Requirements!
- Which ones? Based on which characteristics?
 - ❑ REQUIRES TO DEFINE **RULE APPLICABILITY FOR NEEDS AND REQUIREMENTS!**



Main Content of the Rule Applicability Matrix

- Rule ID / Name / Description
- **RULE APPLICABILITY**
 - CNCR = Compulsory for Needs AND Requirements
 - RNCR = Recommended for Needs; Compulsory for Requirements [+Comments]
 - CR = Compulsory for Requirements (Optional for Needs)
- **RULE SCOPE** : Requires Project Data Dictionary, Glossary, Models or Ontology?
 - Y/N

Main Content of the Rule Applicability Matrix: Example

RULE #	RULE NAME	RULE SHORT DESCRIPTION	RULE APPLICABILITY	APPLICABILITY COMMENTS	RULE SCOPE Requires Project Data Dictionary, Glossary, Models or Ontology? (Y/N)	SCOPE COMMENTS
R01	Sentence Structure	Use the form of a complete sentence: subject, verb, object.	CNCR		N	
R02	Use Active Voice	Use the active voice in the main sentence structure with the responsible entity clearly identified as the subject.	RNCR	 Needs: The use of active voice can have a higher tolerance when considering needs, although it is recommended to avoid it in both cases.	N	

Main Content of the Rule Applicability Matrix: Elaboration

➤ **Rule 01 – Sentence Structure**

- Both Needs and Requirements need a proper, standardized sentence structure
- Project Data Dictionary: Not required

➤ **Rule 02 – Use Active Voice**

- Requirements shall use active voice (identify SOI, express functionality, etc.)
- Needs might use passive voice
- Project Data Dictionary: Not required (generic syntax checking)

Example #1: Needs vs. Requirements

		APPLICABILITY	SCOPE (Requires Project Data Dictionary?)
Rule 02 <i>Use Active Voice</i>		RNCR	No
Rule 05 <i>Use Definite Articles</i>		RNCR	No
Rule 07 <i>Avoid Vague Terms</i>		RNCR	No
Rule 21 <i>Avoid Parentheses</i>		RNCR	No

➤ **Example:** “When invited to enter user credentials, the user shall be notified by a large screen (above 30”) connected to the management system.”

Example #1: Needs vs. Requirements

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font Calibri Font Size 12

When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the management system.

Metric: Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block

N/A

Correctness metrics summary:

Low Quality 0.95

Metric	Value
✓ Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block	1
✓ Accuracy R05 / TRC-M020: Avoid the use of Indefinite Articles	1
✓ Accuracy R07 / TRC-M950: Avoid the use of Vague Terms	1
✓ Singularity R21 / TRC-M390: Avoid the use of Parenthesis out of the condition block	2

[Edit manual assessment](#)

Ready

Save and close Cancel

Example #1: Needs vs. Requirements

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font: Calibri Font Size: 12

When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the management system.

Metric: Accuracy R05 / TRC-M020: Avoid the use of Indefinite Articles

- Term Tag: INDEFINITE ARTICLE
- Scope Note: N/A
- Clusters: N/A

Correctness metrics summary:

Low Quality 0.95

Metric	Value
✓ Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block	1
✓ Accuracy R05 / TRC-M020: Avoid the use of Indefinite Articles	1
✓ Accuracy R07 / TRC-M950: Avoid the use of Vague Terms	1
✓ Singularity R21 / TRC-M390: Avoid the use of Parenthesis out of the condition block	2

[Edit manual assessment](#)

Ready

Save and close Cancel

Example #1: Needs vs. Requirements

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font: Calibri Font Size: 12

When invited to enter user credentials, the user shall be notified by a **large** screen (above 30") connected to the management system.

Metric: Accuracy R07 / TRC-M950: Avoid the use of Vague Terms

- Scope Note: N/A
- Clusters: «REQUIREMENTS» | «VAGUE ADJECTIVES»

Correctness metrics summary:

Low Quality 0.95

Metric	Value
✓ Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block	1
✓ Accuracy R05 / TRC-M020: Avoid the use of Indefinite Articles	1
✓ Accuracy R07 / TRC-M950: Avoid the use of Vague Terms	1
✓ Singularity R21 / TRC-M390: Avoid the use of Parenthesis out of the condition block	2

[Edit manual assessment](#)

Ready

Save and close Cancel

Example #1: Needs vs. Requirements

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font Calibri Font Size 12

When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the management system.

Metric: Singularity R21 / TRC-M390: Avoid the use of Parenthesis out of the condition block

- Scope Note: N/A
- Clusters: «BRACKETS»

Correctness metrics summary:

Low Quality 0.95

Metric	Value
✓ Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block	1
✓ Accuracy R05 / TRC-M020: Avoid the use of Indefinite Articles	1
✓ Accuracy R07 / TRC-M950: Avoid the use of Vague Terms	1
✓ Singularity R21 / TRC-M390: Avoid the use of Parenthesis out of the condition block	2

Checking with rules adapted to requirement statements

[Edit manual assessment](#)

Ready

Save and close Cancel

Example #1: Needs vs. Requirements

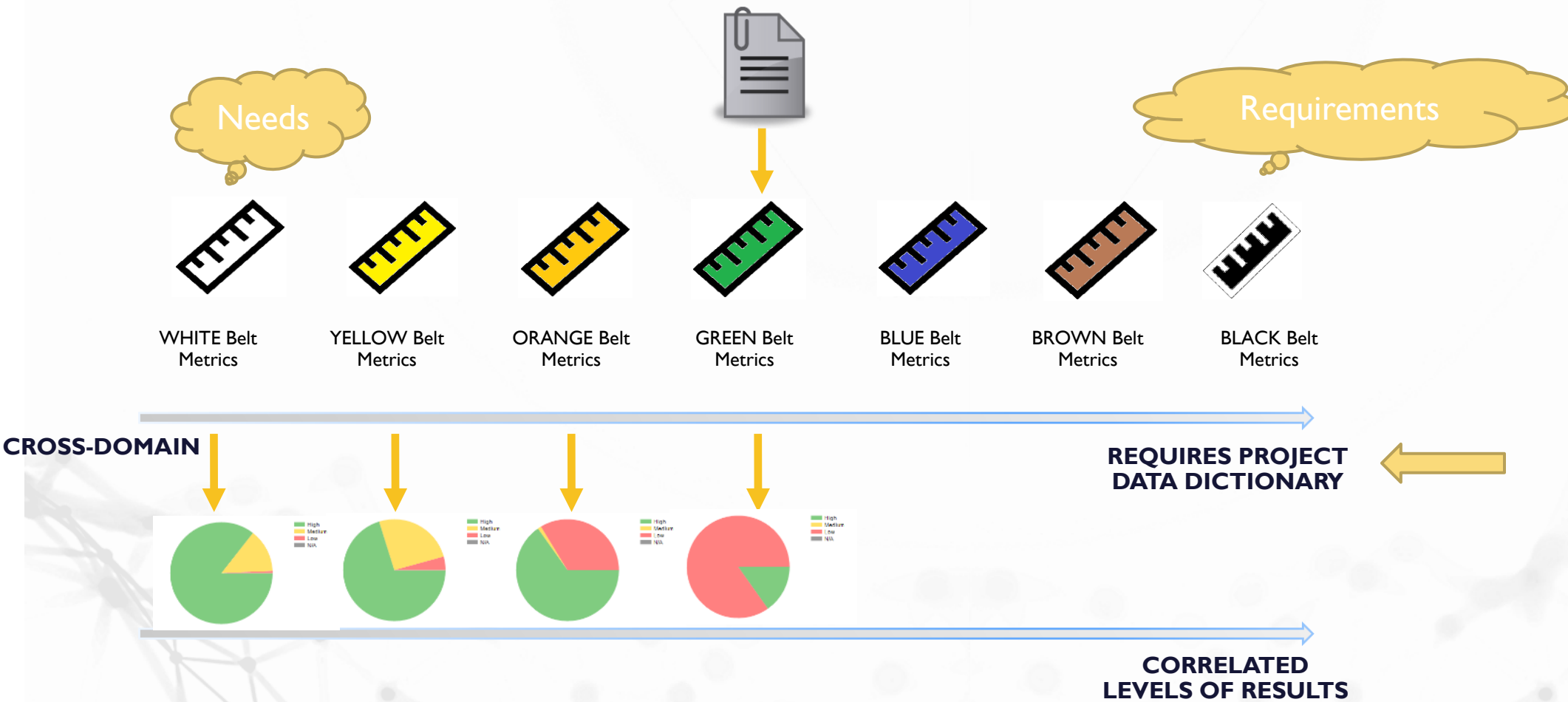
The screenshot displays the SES ENGINEERING Studio interface. The main window is titled "Editing 169 - SES ENGINEERING Studio" and shows a requirement statement: "When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the management system." The text is in Calibri font, size 12. A red message above the text states: "No selected pattern group implies no writing assistance".

On the right side, there is a "Correctness metrics summary" panel. It shows a "High Quality" status with a value of "0.00". Below this, there is a table with the following structure:

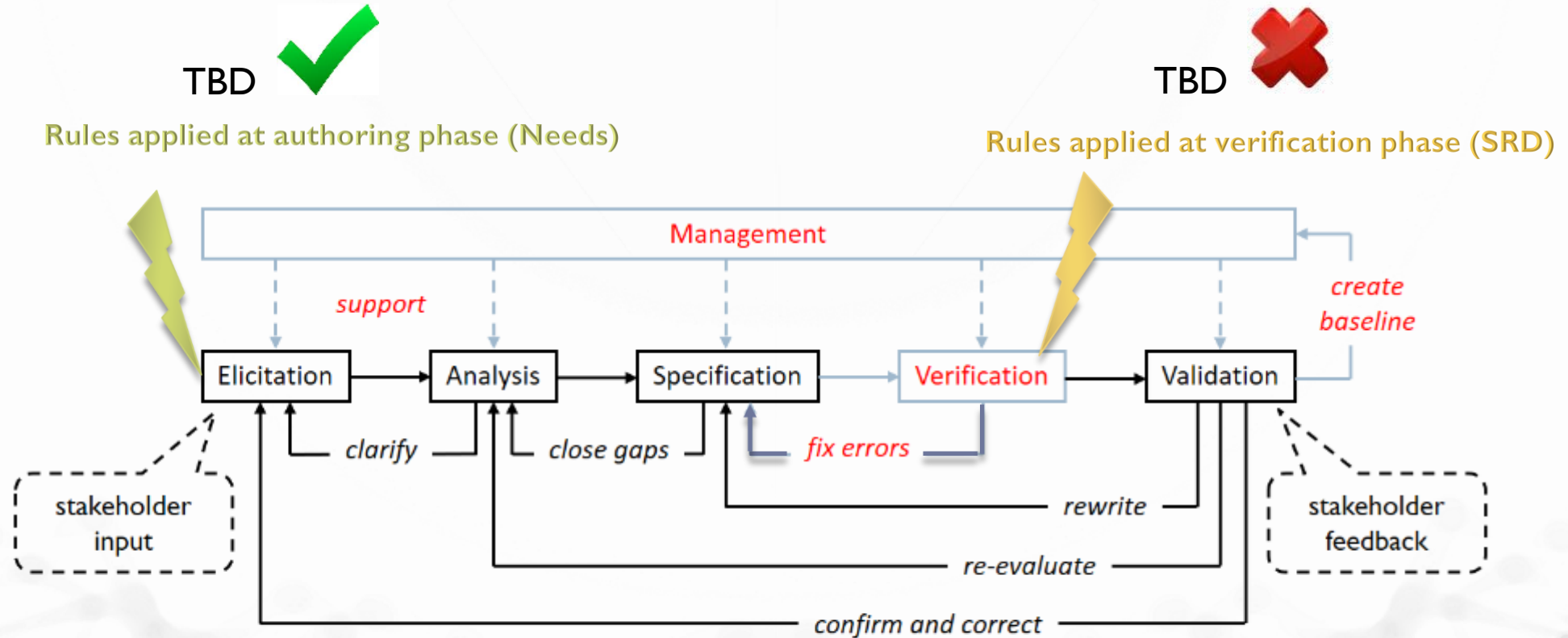
Metric	Value
Checking with rules adapted to need statements	

At the bottom of the panel, there is a link "Edit manual assessment" and a "Ready" status. The bottom of the window has "Save and close" and "Cancel" buttons.

The need for Data Dictionaries

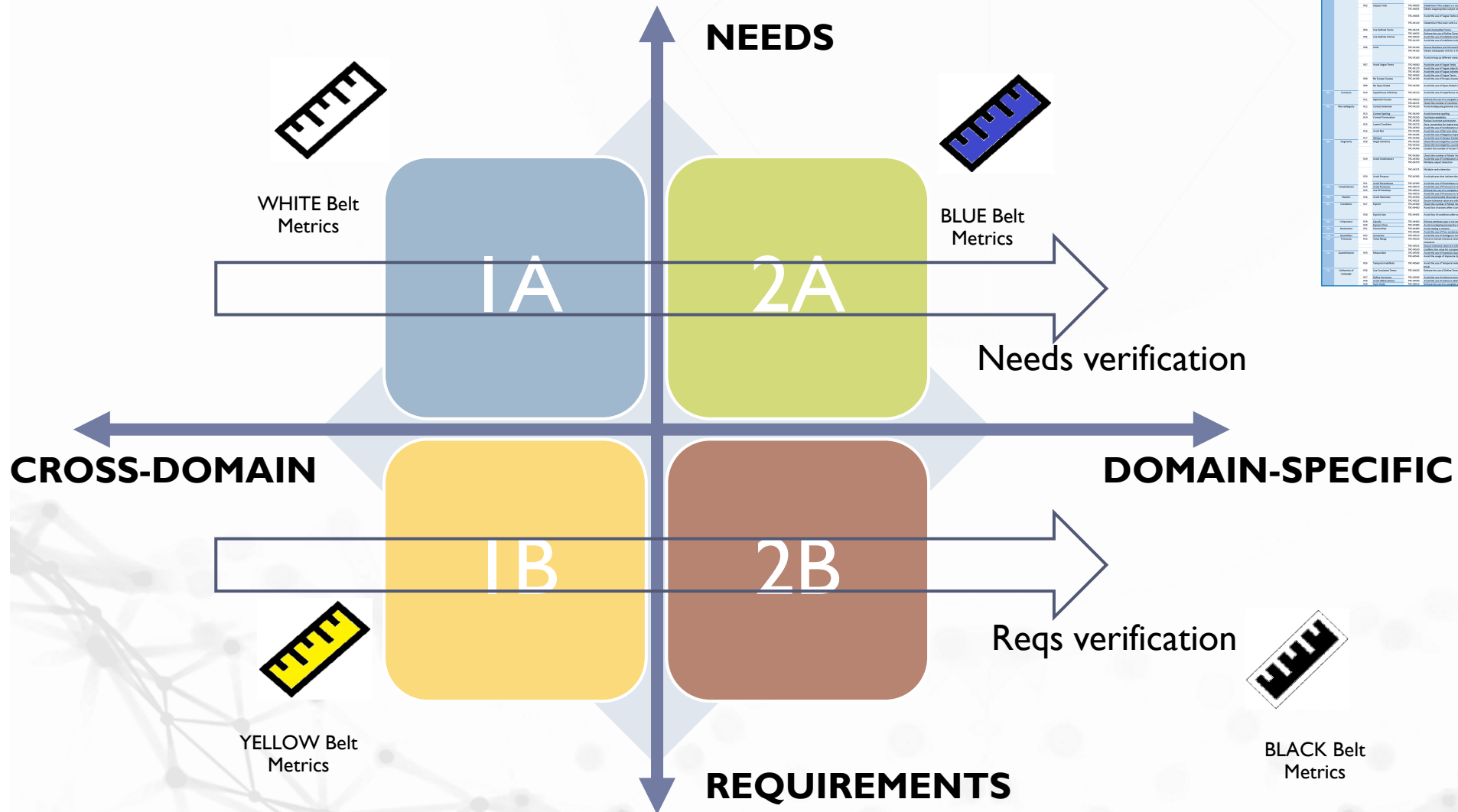


The 3-D tailoring space: Life-cycle Stage – Focus on needs & requirements



Adapted from: Karl Wiegers

INCOSE GtWR with tailoring



Mapping INCOSE 2013 rules with TRC quality metrics

INCOSE 2013 Rule	TRC Quality Metric	Mapping
INCOSE 2013 Rule 1.1.1	TRC Quality Metric 1.1.1	Mapping 1.1.1
INCOSE 2013 Rule 1.1.2	TRC Quality Metric 1.1.2	Mapping 1.1.2
INCOSE 2013 Rule 1.1.3	TRC Quality Metric 1.1.3	Mapping 1.1.3
INCOSE 2013 Rule 1.1.4	TRC Quality Metric 1.1.4	Mapping 1.1.4
INCOSE 2013 Rule 1.1.5	TRC Quality Metric 1.1.5	Mapping 1.1.5
INCOSE 2013 Rule 1.1.6	TRC Quality Metric 1.1.6	Mapping 1.1.6
INCOSE 2013 Rule 1.1.7	TRC Quality Metric 1.1.7	Mapping 1.1.7
INCOSE 2013 Rule 1.1.8	TRC Quality Metric 1.1.8	Mapping 1.1.8
INCOSE 2013 Rule 1.1.9	TRC Quality Metric 1.1.9	Mapping 1.1.9
INCOSE 2013 Rule 1.1.10	TRC Quality Metric 1.1.10	Mapping 1.1.10
INCOSE 2013 Rule 1.1.11	TRC Quality Metric 1.1.11	Mapping 1.1.11
INCOSE 2013 Rule 1.1.12	TRC Quality Metric 1.1.12	Mapping 1.1.12
INCOSE 2013 Rule 1.1.13	TRC Quality Metric 1.1.13	Mapping 1.1.13
INCOSE 2013 Rule 1.1.14	TRC Quality Metric 1.1.14	Mapping 1.1.14
INCOSE 2013 Rule 1.1.15	TRC Quality Metric 1.1.15	Mapping 1.1.15
INCOSE 2013 Rule 1.1.16	TRC Quality Metric 1.1.16	Mapping 1.1.16
INCOSE 2013 Rule 1.1.17	TRC Quality Metric 1.1.17	Mapping 1.1.17
INCOSE 2013 Rule 1.1.18	TRC Quality Metric 1.1.18	Mapping 1.1.18
INCOSE 2013 Rule 1.1.19	TRC Quality Metric 1.1.19	Mapping 1.1.19
INCOSE 2013 Rule 1.1.20	TRC Quality Metric 1.1.20	Mapping 1.1.20
INCOSE 2013 Rule 1.1.21	TRC Quality Metric 1.1.21	Mapping 1.1.21
INCOSE 2013 Rule 1.1.22	TRC Quality Metric 1.1.22	Mapping 1.1.22
INCOSE 2013 Rule 1.1.23	TRC Quality Metric 1.1.23	Mapping 1.1.23
INCOSE 2013 Rule 1.1.24	TRC Quality Metric 1.1.24	Mapping 1.1.24
INCOSE 2013 Rule 1.1.25	TRC Quality Metric 1.1.25	Mapping 1.1.25
INCOSE 2013 Rule 1.1.26	TRC Quality Metric 1.1.26	Mapping 1.1.26
INCOSE 2013 Rule 1.1.27	TRC Quality Metric 1.1.27	Mapping 1.1.27
INCOSE 2013 Rule 1.1.28	TRC Quality Metric 1.1.28	Mapping 1.1.28
INCOSE 2013 Rule 1.1.29	TRC Quality Metric 1.1.29	Mapping 1.1.29
INCOSE 2013 Rule 1.1.30	TRC Quality Metric 1.1.30	Mapping 1.1.30
INCOSE 2013 Rule 1.1.31	TRC Quality Metric 1.1.31	Mapping 1.1.31
INCOSE 2013 Rule 1.1.32	TRC Quality Metric 1.1.32	Mapping 1.1.32
INCOSE 2013 Rule 1.1.33	TRC Quality Metric 1.1.33	Mapping 1.1.33
INCOSE 2013 Rule 1.1.34	TRC Quality Metric 1.1.34	Mapping 1.1.34
INCOSE 2013 Rule 1.1.35	TRC Quality Metric 1.1.35	Mapping 1.1.35
INCOSE 2013 Rule 1.1.36	TRC Quality Metric 1.1.36	Mapping 1.1.36
INCOSE 2013 Rule 1.1.37	TRC Quality Metric 1.1.37	Mapping 1.1.37
INCOSE 2013 Rule 1.1.38	TRC Quality Metric 1.1.38	Mapping 1.1.38
INCOSE 2013 Rule 1.1.39	TRC Quality Metric 1.1.39	Mapping 1.1.39
INCOSE 2013 Rule 1.1.40	TRC Quality Metric 1.1.40	Mapping 1.1.40
INCOSE 2013 Rule 1.1.41	TRC Quality Metric 1.1.41	Mapping 1.1.41
INCOSE 2013 Rule 1.1.42	TRC Quality Metric 1.1.42	Mapping 1.1.42
INCOSE 2013 Rule 1.1.43	TRC Quality Metric 1.1.43	Mapping 1.1.43
INCOSE 2013 Rule 1.1.44	TRC Quality Metric 1.1.44	Mapping 1.1.44
INCOSE 2013 Rule 1.1.45	TRC Quality Metric 1.1.45	Mapping 1.1.45
INCOSE 2013 Rule 1.1.46	TRC Quality Metric 1.1.46	Mapping 1.1.46
INCOSE 2013 Rule 1.1.47	TRC Quality Metric 1.1.47	Mapping 1.1.47
INCOSE 2013 Rule 1.1.48	TRC Quality Metric 1.1.48	Mapping 1.1.48
INCOSE 2013 Rule 1.1.49	TRC Quality Metric 1.1.49	Mapping 1.1.49
INCOSE 2013 Rule 1.1.50	TRC Quality Metric 1.1.50	Mapping 1.1.50
INCOSE 2013 Rule 1.1.51	TRC Quality Metric 1.1.51	Mapping 1.1.51
INCOSE 2013 Rule 1.1.52	TRC Quality Metric 1.1.52	Mapping 1.1.52
INCOSE 2013 Rule 1.1.53	TRC Quality Metric 1.1.53	Mapping 1.1.53
INCOSE 2013 Rule 1.1.54	TRC Quality Metric 1.1.54	Mapping 1.1.54
INCOSE 2013 Rule 1.1.55	TRC Quality Metric 1.1.55	Mapping 1.1.55
INCOSE 2013 Rule 1.1.56	TRC Quality Metric 1.1.56	Mapping 1.1.56
INCOSE 2013 Rule 1.1.57	TRC Quality Metric 1.1.57	Mapping 1.1.57
INCOSE 2013 Rule 1.1.58	TRC Quality Metric 1.1.58	Mapping 1.1.58
INCOSE 2013 Rule 1.1.59	TRC Quality Metric 1.1.59	Mapping 1.1.59
INCOSE 2013 Rule 1.1.60	TRC Quality Metric 1.1.60	Mapping 1.1.60
INCOSE 2013 Rule 1.1.61	TRC Quality Metric 1.1.61	Mapping 1.1.61
INCOSE 2013 Rule 1.1.62	TRC Quality Metric 1.1.62	Mapping 1.1.62
INCOSE 2013 Rule 1.1.63	TRC Quality Metric 1.1.63	Mapping 1.1.63
INCOSE 2013 Rule 1.1.64	TRC Quality Metric 1.1.64	Mapping 1.1.64
INCOSE 2013 Rule 1.1.65	TRC Quality Metric 1.1.65	Mapping 1.1.65
INCOSE 2013 Rule 1.1.66	TRC Quality Metric 1.1.66	Mapping 1.1.66
INCOSE 2013 Rule 1.1.67	TRC Quality Metric 1.1.67	Mapping 1.1.67
INCOSE 2013 Rule 1.1.68	TRC Quality Metric 1.1.68	Mapping 1.1.68
INCOSE 2013 Rule 1.1.69	TRC Quality Metric 1.1.69	Mapping 1.1.69
INCOSE 2013 Rule 1.1.70	TRC Quality Metric 1.1.70	Mapping 1.1.70
INCOSE 2013 Rule 1.1.71	TRC Quality Metric 1.1.71	Mapping 1.1.71
INCOSE 2013 Rule 1.1.72	TRC Quality Metric 1.1.72	Mapping 1.1.72
INCOSE 2013 Rule 1.1.73	TRC Quality Metric 1.1.73	Mapping 1.1.73
INCOSE 2013 Rule 1.1.74	TRC Quality Metric 1.1.74	Mapping 1.1.74
INCOSE 2013 Rule 1.1.75	TRC Quality Metric 1.1.75	Mapping 1.1.75
INCOSE 2013 Rule 1.1.76	TRC Quality Metric 1.1.76	Mapping 1.1.76
INCOSE 2013 Rule 1.1.77	TRC Quality Metric 1.1.77	Mapping 1.1.77
INCOSE 2013 Rule 1.1.78	TRC Quality Metric 1.1.78	Mapping 1.1.78
INCOSE 2013 Rule 1.1.79	TRC Quality Metric 1.1.79	Mapping 1.1.79
INCOSE 2013 Rule 1.1.80	TRC Quality Metric 1.1.80	Mapping 1.1.80
INCOSE 2013 Rule 1.1.81	TRC Quality Metric 1.1.81	Mapping 1.1.81
INCOSE 2013 Rule 1.1.82	TRC Quality Metric 1.1.82	Mapping 1.1.82
INCOSE 2013 Rule 1.1.83	TRC Quality Metric 1.1.83	Mapping 1.1.83
INCOSE 2013 Rule 1.1.84	TRC Quality Metric 1.1.84	Mapping 1.1.84
INCOSE 2013 Rule 1.1.85	TRC Quality Metric 1.1.85	Mapping 1.1.85
INCOSE 2013 Rule 1.1.86	TRC Quality Metric 1.1.86	Mapping 1.1.86
INCOSE 2013 Rule 1.1.87	TRC Quality Metric 1.1.87	Mapping 1.1.87
INCOSE 2013 Rule 1.1.88	TRC Quality Metric 1.1.88	Mapping 1.1.88
INCOSE 2013 Rule 1.1.89	TRC Quality Metric 1.1.89	Mapping 1.1.89
INCOSE 2013 Rule 1.1.90	TRC Quality Metric 1.1.90	Mapping 1.1.90
INCOSE 2013 Rule 1.1.91	TRC Quality Metric 1.1.91	Mapping 1.1.91
INCOSE 2013 Rule 1.1.92	TRC Quality Metric 1.1.92	Mapping 1.1.92
INCOSE 2013 Rule 1.1.93	TRC Quality Metric 1.1.93	Mapping 1.1.93
INCOSE 2013 Rule 1.1.94	TRC Quality Metric 1.1.94	Mapping 1.1.94
INCOSE 2013 Rule 1.1.95	TRC Quality Metric 1.1.95	Mapping 1.1.95
INCOSE 2013 Rule 1.1.96	TRC Quality Metric 1.1.96	Mapping 1.1.96
INCOSE 2013 Rule 1.1.97	TRC Quality Metric 1.1.97	Mapping 1.1.97
INCOSE 2013 Rule 1.1.98	TRC Quality Metric 1.1.98	Mapping 1.1.98
INCOSE 2013 Rule 1.1.99	TRC Quality Metric 1.1.99	Mapping 1.1.99
INCOSE 2013 Rule 1.1.100	TRC Quality Metric 1.1.100	Mapping 1.1.100



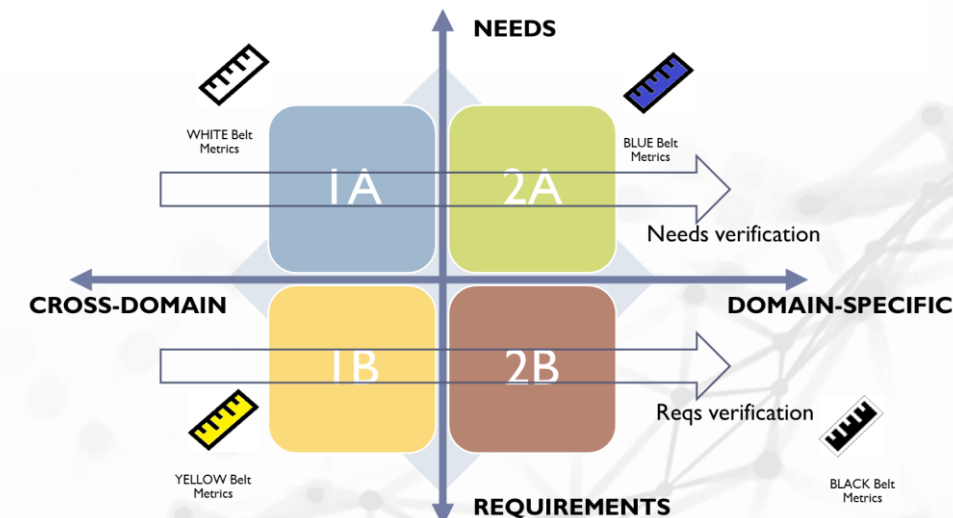
INCOSE GtWR with tailoring

Quality Assurance - Metrics set templates

Metrics set templates:

[+ Add metric set](#)

Identifier	Name	Description
187	1A - INCOSE GtWR v4.0 CROSS-DOMAIN METRICS x NEED Statements	1A. This subset includes cross-domain rules, which do not require domain-specific data dictionaries, and applies to NEED sta
188	1B - INCOSE GtWR v4.0 CROSS-DOMAIN METRICS x REQUIREMENT Statements	1B. This subset includes cross-domain rules, which do not require domain-specific data dictionaries, and applies to REQUIRE
189	2A - INCOSE GtWR v4.0 CROSS-DOMAIN & DOMAIN-SPECIFIC METRICS x NEED Statements	2A. This subset includes cross-domain & domain-specific rules, which require a project data dictionary, and applies to NEED
190	2B - INCOSE GtWR v4.0 CROSS-DOMAIN & DOMAIN-SPECIFIC METRICS x REQUIREMENT Statements	2B. This subset includes cross-domain & domain-specific rules, which require a project data dictionary, and applies to REQU



INCOSE GtWR with tailoring

Metrics set template configuration: 1B - INCOSE GtWR v4.0 CROSS-DOMAIN METRICS x REQUIREMENT Statements

Metrics configuration:

Correctness Consistency Completeness

Correctness metrics:

Search

+ Add new Edit Delete

Metric Identifier	Custom Metric...	Name	Rationale	Weight	En...	Correctness type
32,401	N/A	Abstraction R31 / TRC-M490: Avoid stating a solution	This quality met...	1	☒	Parameterized - Cluster
32,402	N/A	Abstraction R31 / TRC-M500: Avoid the use of Flow sentences	This metric chec...	1	☒	Parameterized - Cluster
32,365	N/A	Accuracy R01 / TRC-M365: Avoid the use of Banned Modal Verbs	This is a metric...	1	☒	Parameterized - Cluster
32,367	N/A	Accuracy R02 / TRC-M040: Avoid the use of Passive Voice out of the condition block	This metric chec...	1	☒	Parameterized - Pattern matching
32,369	N/A	Accuracy R05 / TRC-M130: Avoid the use of Indefinite Articles in front of an Agent	This metric chec...	1	☒	Parameterized - Pattern matching
32,372	N/A	Accuracy R06 / TRC-M140: Ensure Numbers are followed by Units or noun qualifications	This metric chec...	1	☒	Parameterized - Term tag
32,371	N/A	Accuracy R06 / TRC-M150: Detect inadequate Unit for a Characteristic	hec...	1	☒	Parameterized - Relationships not SCM comp...
32,376	N/A	Accuracy R07 / TRC-M950: Avoid the use of Vague Terms	hec...	1	☒	Parameterized - Cluster
32,377	N/A	Accuracy R08 / TRC-M190: Avoid the use of Escape clauses	hec...	1	☒	Parameterized - Special Sentences
32,378	N/A	Accuracy R09 / TRC-M200: Avoid the use of Open-Ended clauses	hec...	1	☒	Parameterized - Special Sentences
32,397	N/A	Completeness R24 / Completeness R25 / TRC-M070: Avoid the use of Pronouns t	met...	1	☒	Parameterized - Term tag
32,379	N/A	Concision R10 / TRC-M210: Avoid the use of Superfluous infinitives	hec...	1	☒	Parameterized - Pattern group matching
32,380	N/A	Concision R11 / TRC-M215: Check the number of condition clauses	me...	1	☒	Parameterized - Cluster
32,381	N/A	Non-ambiguity R12 / TRC-M230: Avoid incorrect grammar structures	cor...	1	☒	Parameterized - Pattern group matching
32,383	N/A	Non-ambiguity R14 / TRC-M250: Facilitate readability	ba...	1	☒	Non-parameterized
32,382	N/A	Non-ambiguity R14 / TRC-M260: Review incorrect punctuation	hec...	1	☒	Non-parameterized
32,384	N/A	Non-ambiguity R15 / Singularity R19 / TRC-M350: Avoid the use of Combinators	hec...	1	☒	Parameterized - Cluster
32,385	N/A	Non-ambiguity R15 / TRC-M270: Set a convention for logical expression forms	hec...	1	☒	Parameterized - Cluster

No. of metrics: 38, Enabled: 38

☒ Enabled

- + Add new metric
- Edit metric
- Delete metric(s)
- Enable Selected
- Enable all
- Disable all
- View text as paragraphs
- Search...
- Copy to
- Select all
- Select none
- Invert selection
- Copy selection to clipboard
- Export...
- Refresh

Why do we need a Project Data Dictionary?

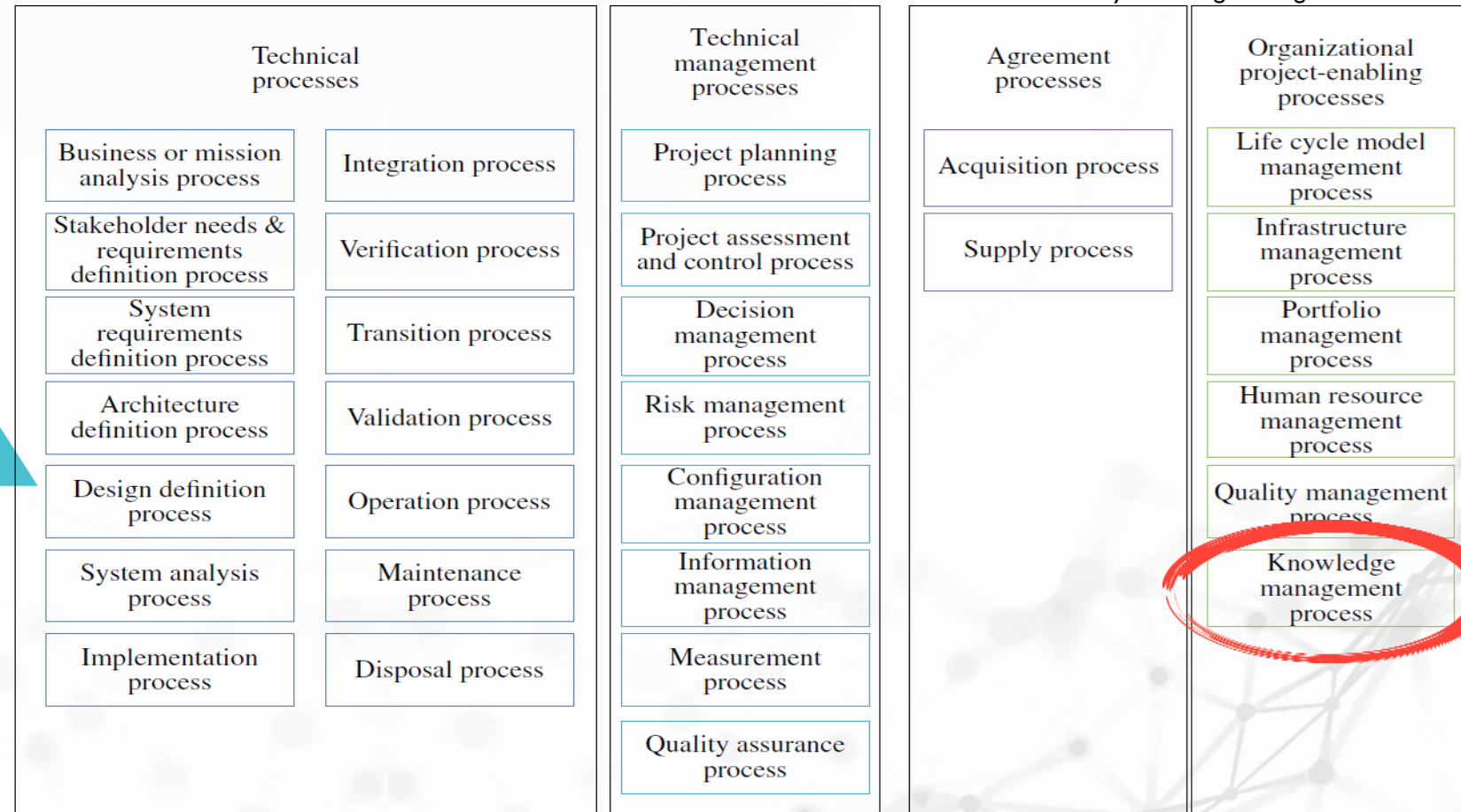
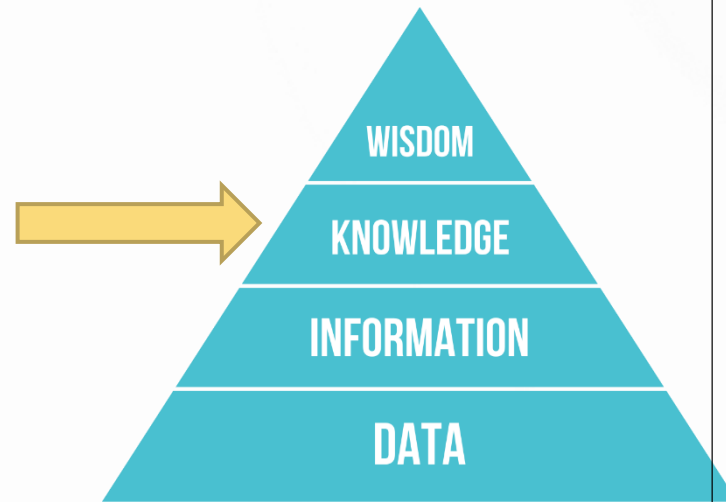
“centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format” (IBM Dictionary of Computing, 1993)

“Agreed-upon set of standards and guidelines to enforce the use of common vocabulary and concepts, thus ensuring the consistency of Needs and Requirements across the project team”

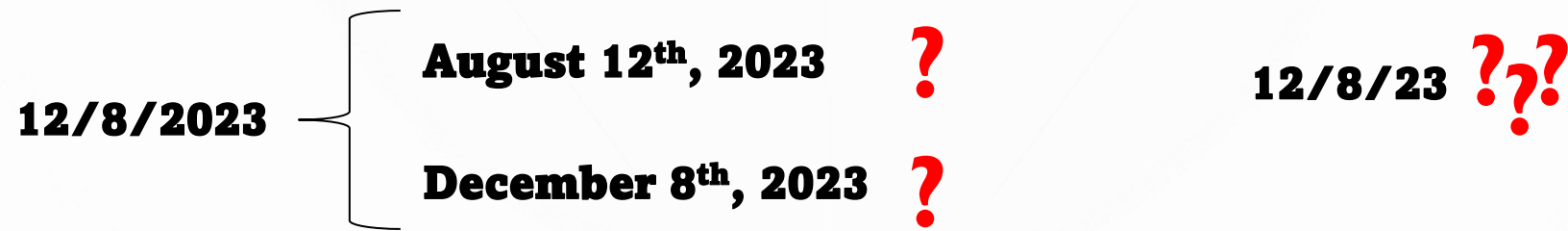
- **Consistent** wording (pre-established by the subject matter experts)
- **Extended** coverage of the INCOSE GtWR set of rules (various rules require a dictionary to be assessed).
- **Standardized** way of drafting requirements : better traceability (either manual or based on NLP), reusability and avoided redundancy (duplicated or contradictory requirements).

Why do we need a Project Data Dictionary?

➤ Data -> Knowledge



Why do we need a Project Data Dictionary?

**WHO**

who.int

OMS

- Let's **imagine the ambiguity** that might be introduced while dealing with **Complex Systems!!**

Example #2: Applying Project Data Dictionaries

	APPLICABILITY	SCOPE (Requires Project Data Dictionary?)
Rule 03 <i>Subject Verb</i>	CNCR	X
Rule 04 <i>Use Defined Terms</i>	CNCR	X
Rule 13 <i>Correct Spelling</i>	CNCR	X
Rule 37 <i>Define Acronym</i>	CNCR	X

➤ **Example:** “While in Combat Mode, The **TW**a shall Receive Attacks”

- Rule 03: Invalid subject (not known as a system, entity, stakeholder, function, activity, etc.)
- Rule 04: Undefined term (not defined in the glossary)
- Rule 13: Detection as misspelled word
- Rule 37: Undefined acronym

Example #2: Applying Project Data Dictionaries

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font: Calibri Font Size: 12

While in Combat Mode, The TWA shall Receive Attacks.

Metric: Uniformity of Language R37 / TRC-M580: Avoid the use of unknown acronyms

N/A

Correctness metrics summary:

Low Quality 0.71

Metric	Value
✓ Accuracy R03 / TRC-M050: Determine if the subject is a recognized Agent term	0
✓ Accuracy R04 / TRC-M225: Avoid Unclassified Terms	1
✓ Uniformity of Language R37 / TRC-M580: Avoid the use of unknown acronyms	1

[Edit manual assessment](#)

Ready

Save and close Cancel

Example #2: Applying Project Data Dictionaries

The screenshot displays the SES ENGINEERING Studio interface. The main window is titled "Editing 169 - SES ENGINEERING Studio" and shows a requirement statement: "When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the management system." The interface includes a menu bar (File, Tools, View, Log) and a toolbar with various editing tools. A dropdown menu shows "< No pattern group >". A red message states: "No selected pattern group implies no writing assistance".

On the right side, a "Correctness metrics summary" panel is visible, showing a "High Quality" status with a value of "0.00". The panel includes a table with columns "Metric" and "Value". Below the table, the text "Checking with rules adapted to need statements" is displayed in green. At the bottom of the panel, there is a link "Edit manual assessment" and a "Ready" status. The bottom of the window features "Save and close" and "Cancel" buttons.

Example #2: Applying Project Data Dictionaries

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing

Font Calibri Font Size 12

When invited to enter **user credentials**, the user shall be notified by a large screen (above 30") connected to the management system.

[SysR-154] The Temperature Warrior shall Configure the Round parameters.

Loaded SysR

Term - Read Only Mode

Term configuration:

Identifier: 61,873 Name: User credentials

Belongs to Domain: ☒ Ignore accents (diacritics) exception: ☐ Keep the original format of the term: ☐

Syntactic and semantic configuration:

Term tag: NOUN

Cluster(s): 0 cluster(s)

Relationship type:

Language: English (United Kingdom)

Gender: N/A Number: Invariant

☒ Changes gender ☒ Changes number

Synchronizes: ☐ Flag 1: ☐ Flag 2: ☐ Flag 3: ☐

Statistics:

TF: 0.000000 DF: 0.000000

TFxDF: 0.000000

Synonyms

SCM Relationships

Documentation

Translations

Documentation information:

Classification code:

Scope note:

User credentials comprise the information required to verify the identity of a user attempting to access a network or computing resources.

History note:

Sources:

OK Cancel

Example #2: Applying Project Data Dictionaries

Editing 169 - SES ENGINEERING Studio

File Tools View Log

Authoring without patterns

< No pattern group >

No selected pattern group implies no writing assistance

Font Calibri Font Size 12

When invited to enter user credentials, the user shall be notified by a large screen (above 30") connected to the **management system**.

Term - Read Only Mode

Term configuration:

Identifier: 61,558 Name: Management system

Belongs to Domain: ☒ Ignore accents (diacritics) exception: ☐ Keep the original format of the term: ☒

Synonyms SCM Relationships

Documentation Translations

Documentation information:

Classification code:

Scope note:

System entity responsible for processing, storing and communicating the information gathered throughout the operation of the system.

History note:

Sources:

Term tag: NOUN

Cluster(s): «SYSTEM ELEMENT»

1 cluster(s)

Relationship type:

Language: English (United Kingdom)

Gender: N/A Number: Invariant

☒ Changes gender ☒ Changes number

Synchronizes: ☐ Flag 1: ☐ Flag 2: ☐ Flag 3: ☐

Statistics:

TF: 0.000000 DF: 0.000000

TFxDF: 0.000000

OK Cancel

[SysR-154] The Temperature Warrior shall Configure the Round parameters.

[SysR-155] The Temperature Warrior shall manage the temperatures

Loaded

SysR-153

SysR-154



6.

**Demo
Rule Applicability
Matrix**

Use-cases

- Use-case #1 : Quality assessment of stakeholder needs
- Use-case #2 : Quality assessment using requirements quality rules + editing
- Use-case #3 : Live quality check + pattern-based writing assistance + data dictionary

SES ENGINEERING Studio

SES Connections Change Management Quality V&V Traceability Interoperability Reports DMS KM

Quality Assurance - Metrics set templates

TW - Stakeholder Needs

Views QUALITY Shared

Quality Assurance Project Configuration

Worksheet Selector Current State Snapshot Evolution Scoreboard Requirements Quality view Users Correctness Consistency Completeness Suggestions

StRS Quality Scoreboard Requirements Metrics Repository

Drag a column header here to group by that column

	C.		Project	Worksheet	ID		Correctness
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-07	Temperature ranges shall involve only one time interval.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-08	Temperature ranges shall involve more than one time interval.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-09	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 30 °C.	★★★☆☆
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-10	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★☆☆
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-11	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-12	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-13	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-14	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-15	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-16	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-17	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-18	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-19	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-20	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-21	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-22	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-23	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-24	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-25	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-26	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-27	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-28	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-29	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-30	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-31	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★
	☐	☐	TW - Stakeholder Requirem...	StRS	SH-32	Temperature ranges shall be allowed for the definition of the temperature ranges shall be 12 °C.	★★★★





Contact information



Ilyes Yousfi



ilyes.yousfi@reusecompany.com



+34 627 08 66 01



@ReuseCompany



<https://www.linkedin.com/in/ilyesyoufsi/en>





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